PADHAKU
Science · Class 10 Science



CHAPTER 11

# Electricity

NCERT Exercise Solutions


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Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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## ❓ Questions · 11.1 Electric Current and Circuit

**Q 1** **What does an electric circuit mean?**

✔ ANSWER

An electric circuit is a **continuous and closed path** along which an electric current flows.

Example: cell → switch → bulb → back to the cell. If the path breaks anywhere, the current stops.

**Q 2** **Define the unit of current.**

✔ ANSWER

The SI unit of current is the **ampere (A)**.

**1 ampere** is the current when **1 coulomb** of charge flows through a cross-section of a conductor in **1 second**: 1 A = 1 C/1 s.

**Q 3** **Calculate the number of electrons constituting one coulomb of charge.**

✔ ANSWER

Given: charge on one electron, $e = 1.6 \times 10^{-19}$ C; total charge Q = 1 C

Formula: Q = n e, so n = Q/e

$n = 1\ C / (1.6 \times 10^{-19}\ C)$

**$n = 6.25 \times 10^{18}$ electrons**

## Questions · 11.2 Electric Potential and Potential Difference

**Q 1** **Name a device that helps to maintain a potential difference across a conductor.**

✔ ANSWER

A **cell or a battery** (made of one or more cells). Its chemical action keeps a potential difference across its terminals.

**Q 2** **What is meant by saying that the potential difference between two points is 1 V?**

✔ ANSWER

It means that **1 joule of work** is done to move a charge of **1 coulomb** from one point to the other.

$1\ V = 1\ J/1\ C = 1\ J\ C^{-1}$

**Q 3** **How much energy is given to each coulomb of charge passing through a 6 V battery?**

✔ ANSWER

Given: V = 6 V, Q = 1 C

Formula: $W = V \times Q$

$W = 6\ V \times 1\ C$

**W = 6 J** (each coulomb gets 6 joules of energy)

## ⭐ Activities · 11.4 Ohm's Law

**Activity 11.1** **Connect a nichrome wire XY, an ammeter, a voltmeter and 1, 2, 3, then 4 cells of 1.5 V. Note I and V each time, find V/I and plot a V–I graph. What is the nature of the graph?**

✔ ANSWER

**Observation:** as the number of cells increases, both the current I and the potential difference V increase. The ratio V/I comes out **almost the same** in every case.

The V–I graph is a **straight line passing through the origin**.

**Conclusion:** $V \propto I$, i.e. $V/I$ = constant = R (resistance of the wire). This is **Ohm's law**: $V = IR$, provided the temperature stays the same.

**Activity 11.2** **Connect a nichrome wire, then a torch bulb, then a 10 W bulb in the gap XY of a circuit with four 1.5 V cells and an ammeter. Are the ammeter readings different for different components? What do the observations indicate?**

✔ ANSWER

**Observation:** yes, the ammeter reading is **different** for each component, though the cells (voltage) are the same.

**Conclusion:** different components offer **different resistance** to the flow of current. A component with low resistance lets more current flow; one with high resistance lets less current flow.

## ⭐ Activities · 11.5 Factors on which the Resistance of a Conductor Depends

**Activity 11.3** **Use nichrome wires of length l, of length 2l, a thicker one of length l, and a copper wire of the same length and thickness as the first. Does the current depend on the length of the conductor? Does it depend on the area of cross-section?**

✔ ANSWER

**Observations:**

- Wire of length 2l: current becomes **half** → resistance doubles.
- Thicker wire (same length): current **increases** → resistance decreases.
- Copper wire of same size: current is **different** (more) → resistance depends on the material.

**Yes**, current depends on length (longer wire, less current) and on area of cross-section (thicker wire, more current).

**Conclusion:** $R \propto l$ and $R \propto 1/A$, so $\mathbf{R = \rho\, l/A}$, where ρ is the resistivity of the material.

## Questions · 11.5 Factors on which the Resistance of a Conductor Depends

**Q 1** **On what factors does the resistance of a conductor depend?**

✔ ANSWER

The resistance of a conductor depends on:

- its **length** ($R \propto l$)
- its **area of cross-section** ($R \propto 1/A$)
- the **nature of its material** (resistivity $\rho$)
- its **temperature**

So, $R = \rho\, l/A$.

**Q 2** **Will current flow more easily through a thick wire or a thin wire of the same material, when connected to the same source? Why?**

✔ ANSWER

Current flows more easily through the **thick wire**.

Reason: $R = \rho\, l/A$. A thick wire has a larger area of cross-section A, so its **resistance is less**, and more current flows.

**Q 3** **The resistance of an electrical component remains constant while the potential difference across its ends decreases to half of its former value. What change will occur in the current through it?**

✔ ANSWER

By Ohm's law, $I = V/R$.

If R stays the same and V becomes V/2, then new current $I' = (V/2)/R = I/2$.

**The current also becomes half.**

**Q 4** **Why are coils of electric toasters and electric irons made of an alloy rather than a pure metal?**

✔ ANSWER

- An alloy has **higher resistivity** than its pure metals, so it produces more heat.
- An alloy **does not oxidise (burn) easily** at high temperatures.

**Q 5** **Use the data in Table 11.2 to answer: (a) Which among iron and mercury is a better conductor? (b) Which material is the best conductor?**

✔ ANSWER

(a) **Iron** – its resistivity ($10.0 \times 10^{-8}$ Ω m) is lower than that of mercury ($94.0 \times 10^{-8}$ Ω m).

(b) **Silver** – it has the lowest resistivity ($1.60 \times 10^{-8}$ Ω m) in the table.

## ⭐ Activities · 11.6.1 Resistors in Series

**Activity 11.4** **Join three resistors (e.g. 1 Ω, 2 Ω, 3 Ω) in series with a 6 V battery, an ammeter and a plug key. Move the ammeter to different places between the resistors. Do you find any change in the current?**

✔ ANSWER

**Observation: no change** – the ammeter shows the same reading at every position.

**Conclusion:** in a series combination, the **same current** flows through every part of the circuit (through each resistor).

**Activity 11.5** **Measure the potential difference V across the whole series combination and $V_1$, $V_2$, $V_3$ across each resistor. Deduce a relationship between V, $V_1$, $V_2$ and $V_3$.**

✔ ANSWER

**Observation:** the voltmeter reading across the combination is equal to the p.d. across the battery terminals, and $V_1 + V_2 + V_3$ equals V.

**Relationship: $V = V_1 + V_2 + V_3$**

**Conclusion:** using V = IR for each resistor, the equivalent resistance in series is $R_s = R_1 + R_2 + R_3$.

## ❓ Questions · 11.6.1 Resistors in Series

**Q 1** **Draw a schematic diagram of a circuit consisting of a battery of three cells of 2 V each, a 5 Ω resistor, an 8 Ω resistor, and a 12 Ω resistor, and a plug key, all connected in series.**

💡 SAMPLE ANSWER

Draw in one closed loop (series):

- a **battery of three cells** joined end to end (+ of one to − of the next), total 6 V
- then a **plug key**
- then the **5 Ω**, **8 Ω** and **12 Ω** resistors one after another (zig-zag symbols)
- the wire from the last resistor goes back to the other terminal of the battery.

Use the symbols of Table 11.1. (Draw the diagram in your notebook.)

**Q 2** **Redraw the circuit of Question 1, putting in an ammeter to measure the current through the resistors and a voltmeter to measure the potential difference across the 12 Ω resistor. What would be the readings in the ammeter and the voltmeter?**

✔ ANSWER

**Diagram:** connect the ammeter **in series** in the loop; connect the voltmeter **in parallel** across the ends of the 12 Ω resistor.

Given: V = 3 × 2 V = 6 V; R = 5 + 8 + 12 = 25 Ω

Ammeter: I = V/R = 6 V/25 Ω = **0.24 A**

Voltmeter: $V_{12}$ = I × R = 0.24 A × 12 Ω = **2.88 V**

## ⭐ Activities · 11.6.2 Resistors in Parallel

**Activity 11.6** **Connect three resistors $R_1$, $R_2$, $R_3$ in parallel with a battery, plug key and ammeter (total current I). Then measure the current through each resistor ($I_1$, $I_2$, $I_3$). What is the relationship between I, $I_1$, $I_2$ and $I_3$?**

✔ ANSWER

**Observation:** the voltmeter shows the **same potential difference V** across each resistor. The total current is the sum of the branch currents.

**Relationship: $I = I_1 + I_2 + I_3$**

**Conclusion:** using $I = V/R$, the equivalent resistance in parallel is given by $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$.

## ❓ Questions · 11.6.2 Resistors in Parallel

**Q 1** **Judge the equivalent resistance when the following are connected in parallel – (a) 1 Ω and $10^6$ Ω, (b) 1 Ω and $10^3$ Ω, and $10^6$ Ω.**

✔ ANSWER

In parallel, the equivalent resistance is **less than the smallest** resistance.

(a) $1/R_p = 1/1 + 1/10^6 = 1.000001 \rightarrow R_p \approx 0.999999$ Ω, i.e. **slightly less than 1 Ω (≈ 1 Ω)**

(b) $1/R_p = 1/1 + 1/10^3 + 1/10^6 = 1.001001 \rightarrow R_p \approx$ **0.999 Ω, slightly less than 1 Ω**

**Q 2** **An electric lamp of 100 Ω, a toaster of 50 Ω and a water filter of 500 Ω are connected in parallel to a 220 V source. What is the resistance of an electric iron connected to the same source that takes as much current as all three appliances, and what is the current through it?**

✔ ANSWER

Given: $R_1 = 100$ Ω, $R_2 = 50$ Ω, $R_3 = 500$ Ω, $V = 220$ V

$1/R_p = 1/100 + 1/50 + 1/500 = (5 + 10 + 1)/500 = 16/500$

$R_p$ = 500/16 = **31.25 Ω** (resistance of the iron)

Current: I = V/R = 220 V/31.25 Ω = **7.04 A**

**Q 3** **What are the advantages of connecting electrical devices in parallel with the battery instead of connecting them in series?**

✔ ANSWER

- Each device gets the **same (full) voltage** of the source.
- Each device draws the **current it needs**, depending on its own resistance.
- If one device fails or is switched off, the **others keep working**.
- The total resistance is lower, and each device can have its own switch.

**Q 4** **How can three resistors of resistances 2 Ω, 3 Ω, and 6 Ω be connected to give a total resistance of (a) 4 Ω, (b) 1 Ω?**

✔ ANSWER

(a) Connect **3 Ω and 6 Ω in parallel**, and this combination **in series with 2 Ω**:

1/R = 1/3 + 1/6 = 3/6 → R = 2 Ω; total = 2 Ω + 2 Ω = **4 Ω**

(b) Connect **all three in parallel**:

1/R = 1/2 + 1/3 + 1/6 = (3 + 2 + 1)/6 = 1 → **R = 1 Ω**

**Q 5** **What is (a) the highest, (b) the lowest total resistance that can be secured by combinations of four coils of resistance 4 Ω, 8 Ω, 12 Ω, 24 Ω?**

✔ ANSWER

(a) Highest: all in **series**: R = 4 + 8 + 12 + 24 = **48 Ω**

(b) Lowest: all in **parallel**:

1/R = 1/4 + 1/8 + 1/12 + 1/24 = (6 + 3 + 2 + 1)/24 = 12/24 = 1/2

**R = 2 Ω**

## ? Questions · 11.7 Heating Effect of Electric Current

**Q 1** **Why does the cord of an electric heater not glow while the heating element does?**

✔ ANSWER

The cord is made of **copper/aluminium**, which has **very low resistance**, so very little heat is produced in it ($H = I^2Rt$).

The heating element is made of an alloy like **nichrome**, which has **high resistance**, so a lot of heat is produced and it becomes red hot and glows.

**Q 2** **Compute the heat generated while transferring 96000 coulomb of charge in**

**one hour through a potential difference of 50 V.**

✔ ANSWER

Given: Q = 96000 C, V = 50 V, t = 1 h

Formula: H = VIt = VQ

H = 50 V × 96000 C

**H = 4.8 × $10^6$ J**

**Q 3** **An electric iron of resistance 20 Ω takes a current of 5 A. Calculate the heat developed in 30 s.**

✔ ANSWER

Given: R = 20 Ω, I = 5 A, t = 30 s

Formula: H = $I^2Rt$

H = $(5\ A)^2$ × 20 Ω × 30 s = 25 × 20 × 30

**H = 15000 J = 1.5 × $10^4$ J**

## ❓ Questions · 11.8 Electric Power

**Q 1** **What determines the rate at which energy is delivered by a current?**

✔ ANSWER

**Electric power** determines it. Power is the rate at which electrical energy is consumed or delivered:

P = VI = $I^2R$ = $V^2/R$

**Q 2** **An electric motor takes 5 A from a 220 V line. Determine the power of the motor and the energy consumed in 2 h.**

✔ ANSWER

Given: I = 5 A, V = 220 V, t = 2 h

Power: P = VI = 220 V × 5 A = **1100 W (1.1 kW)**

Energy: E = P × t = 1100 W × 2 h = 2200 W h

**E = 2.2 kW h = 7.92 × $10^6$ J**

## 📝 Exercises

**1** **A piece of wire of resistance R is cut into five equal parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R′, then the ratio R/R′ is –**
**(a) 1/25 (b) 1/5 (c) 5 (d) 25**

✔ ANSWER

**(d) 25**

Each part = R/5. Five in parallel: 1/R′ = 5 × 5/R = 25/R, so R′ = R/25 and R/R′ = 25.

**2** **Which of the following terms does not represent electrical power in a circuit?**
**(a) $I^2R$ (b) $IR^2$ (c) VI (d) $V^2/R$**

✔ ANSWER

**(b) $IR^2$**

Power is $P = VI = I^2R = V^2/R$; $IR^2$ is not a formula for power.

**3** **An electric bulb is rated 220 V and 100 W. When it is operated on 110 V, the power consumed will be –**
**(a) 100 W (b) 75 W (c) 50 W (d) 25 W**

✔ ANSWER

**(d) 25 W**

$R = V^2/P = (220)^2/100 = 484\ \Omega$

At 110 V: $P = V^2/R = (110)^2/484 = 12100/484 = 25$ W

**4** **Two conducting wires of the same material and of equal lengths and equal diameters are first connected in series and then parallel in a circuit across the same potential difference. The ratio of heat produced in series and parallel combinations would be –**
**(a) 1:2 (b) 2:1 (c) 1:4 (d) 4:1**

✔ ANSWER

**(c) 1:4**

Let each wire have resistance R. Series: $R_s = 2R$; parallel: $R_p = R/2$.

For the same V and t, $H = V^2t/R$, so $H_s/H_p = R_p/R_s = (R/2)/(2R) = 1/4$.

**5** **How is a voltmeter connected in the circuit to measure the potential difference between two points?**

✔ ANSWER

A voltmeter is always connected **in parallel** across the two points between which the potential difference is to be measured.

Its + terminal goes to the point at higher potential (towards the + side of the cell).

**6** **A copper wire has diameter 0.5 mm and resistivity of $1.6 \times 10^{-8}$ Ω m. What will be the length of this wire to make its resistance 10 Ω? How much does the resistance change if the diameter is doubled?**

✔ ANSWER

Given: d = 0.5 mm = $5 \times 10^{-4}$ m, r = $2.5 \times 10^{-4}$ m, ρ = $1.6 \times 10^{-8}$ Ω m, R = 10 Ω

Area A = $\pi r^2 = 3.14 \times (2.5 \times 10^{-4})^2 = 1.96 \times 10^{-7}$ $m^2$

Formula: R = ρl/A, so l = RA/ρ

l = $(10 \times 1.96 \times 10^{-7})/(1.6 \times 10^{-8})$

**l ≈ 122.7 m**

If the diameter is doubled, area becomes 4 times ($A \propto d^2$), so resistance becomes **one-fourth**:

New R = 10 Ω/4 = **2.5 Ω** (it decreases by 7.5 Ω).

**7** **The values of current I in a resistor for the values of potential difference V are given below. Plot a graph between V and I and calculate the resistance of that resistor.**

| I (amperes) | 0.5 | 1.0 | 2.0 | 3.0 | 4.0 |
|---|---|---|---|---|---|
| V (volts) | 1.6 | 3.4 | 6.7 | 10.2 | 13.2 |

✔ ANSWER

**Graph:** take I on the x-axis and V on the y-axis, plot the five points. They lie (nearly) on a **straight line through the origin**.

Resistance = slope of the V–I graph:

$R = \Delta V/\Delta I = (13.2 - 1.6)\ V/(4.0 - 0.5)\ A = 11.6/3.5$

**$R \approx 3.3\ \Omega$**

(Check: V/I for each pair is 3.2, 3.4, 3.35, 3.4, 3.3 – about 3.3 Ω.)

**8** **When a 12 V battery is connected across an unknown resistor, there is a current of 2.5 mA in the circuit. Find the value of the resistance of the resistor.**

✔ ANSWER

Given: V = 12 V, I = 2.5 mA = $2.5 \times 10^{-3}$ A

Formula: R = V/I

$R = 12\ V/(2.5 \times 10^{-3}\ A)$

**$R = 4800\ \Omega = 4.8\ k\Omega$**

**9** **A battery of 9 V is connected in series with resistors of 0.2 Ω, 0.3 Ω, 0.4 Ω, 0.5 Ω and 12 Ω, respectively. How much current would flow through the 12 Ω resistor?**

✔ ANSWER

Given: V = 9 V

Total resistance (series): R = 0.2 + 0.3 + 0.4 + 0.5 + 12 = 13.4 Ω

I = V/R = 9 V/13.4 Ω

**I ≈ 0.67 A**

In series the same current flows through every resistor, so 0.67 A flows through the 12 Ω resistor.

**10** **How many 176 Ω resistors (in parallel) are required to carry 5 A on a 220 V line?**

✔ ANSWER

Given: V = 220 V, I = 5 A, each R = 176 Ω

Required total resistance: $R_p$ = V/I = 220/5 = 44 Ω

For n equal resistors in parallel: $R_p$ = R/n

n = $R/R_p$ = 176/44

**n = 4 resistors**

**11** **Show how you would connect three resistors, each of resistance 6 Ω, so that the combination has a resistance of (i) 9 Ω, (ii) 4 Ω.**

✔ ANSWER

(i) Connect **two resistors in parallel** and this pair **in series with the third**:

Parallel pair: 1/R = 1/6 + 1/6 → R = 3 Ω; total = 3 + 6 = **9 Ω**

(ii) Connect **two resistors in series** and this pair **in parallel with the third**:

Series pair = 12 Ω; 1/R = 1/12 + 1/6 = 3/12 → **R = 4 Ω**

**12** **Several electric bulbs designed to be used on a 220 V electric supply line are rated 10 W. How many lamps can be connected in parallel with each other across the two wires of 220 V line if the maximum allowable current is 5 A?**

✔ ANSWER

Given: V = 220 V, power of each bulb = 10 W, maximum current = 5 A

Maximum total power: P = VI = 220 V × 5 A = 1100 W

Number of bulbs = 1100 W/10 W

**110 lamps**

**13** **A hot plate of an electric oven connected to a 220 V line has two resistance coils A and B, each of 24 Ω resistance, which may be used separately, in series, or in parallel. What are the currents in the three cases?**

✔ ANSWER

Given: V = 220 V, R_A = R_B = 24 Ω

**Separately:** I = 220/24 ≈ **9.17 A**

**In series:** R = 24 + 24 = 48 Ω; I = 220/48 ≈ **4.58 A**

**In parallel:** R = 24/2 = 12 Ω; I = 220/12 ≈ **18.33 A**

**14** **Compare the power used in the 2 Ω resistor in each of the following circuits: (i) a 6 V battery in series with 1 Ω and 2 Ω resistors, and (ii) a 4 V battery in parallel with 12 Ω and 2 Ω resistors.**

✔ ANSWER

(i) Total R = 1 + 2 = 3 Ω; I = 6 V/3 Ω = 2 A

Power in 2 Ω: P = $I^2R$ = $(2)^2 \times 2$ = **8 W**

(ii) In parallel, the p.d. across the 2 Ω resistor = 4 V

P = $V^2/R$ = $(4)^2/2$ = **8 W**

**The power used in the 2 Ω resistor is the same (8 W) in both circuits.**

**15** **Two lamps, one rated 100 W at 220 V, and the other 60 W at 220 V, are connected in parallel to electric mains supply. What current is drawn from the line if the supply voltage is 220 V?**

✔ ANSWER

In parallel, each lamp gets 220 V, so both work at their rated power.

Total power P = 100 W + 60 W = 160 W

I = P/V = 160 W/220 V

**I ≈ 0.73 A**

**16** **Which uses more energy, a 250 W TV set in 1 hr, or a 1200 W toaster in 10 minutes?**

✔ ANSWER

Energy E = P × t

TV: E = 250 W × 3600 s = $9 \times 10^5$ J (0.25 kW h)

Toaster: E = 1200 W × 600 s = $7.2 \times 10^5$ J (0.2 kW h)

**The TV set uses more energy.**

**17** **An electric heater of resistance 44 Ω draws 5 A from the service mains for 2 hours. Calculate the rate at which heat is developed in the heater.**

✔ ANSWER

Rate of heat production = power

Given: $R = 44\ \Omega$, $I = 5\ A$

$P = I^2R = (5)^2 \times 44$

**$P = 1100\ W\ (1100\ J/s)$**

(Total heat in 2 h = $1100 \times 7200 = 7.92 \times 10^6$ J.)

**18 (a)** **Why is tungsten used almost exclusively for filament of electric lamps?**

✔ ANSWER

- Tungsten has a **very high melting point (3380°C)**, so it does not melt when it becomes white hot.
- It is strong, has high resistivity, and glows (emits light) at high temperature without melting.

**18 (b)** **Why are the conductors of electric heating devices, such as bread-toasters and electric irons, made of an alloy rather than a pure metal?**

✔ ANSWER

- An alloy has **higher resistivity** than a pure metal, so it produces more heat.
- An alloy **does not oxidise (burn) readily** at high temperatures.

**18 (c)** **Why is the series arrangement not used for domestic circuits?**

✔ ANSWER

- In series, the **same current** flows through all appliances, but appliances need different currents.
- The voltage is shared, so each appliance does not get the full 220 V.
- If one appliance fails or is switched off, the **circuit breaks and all others stop**.
- The total resistance becomes large, so the current becomes small.

**18 (d)** **How does the resistance of a wire vary with its area of cross-section?**

✔ ANSWER

Resistance is **inversely proportional** to the area of cross-section: $R \propto 1/A$.

A thicker wire (larger A) has less resistance; if A is doubled, R becomes half.

**18 (e)** **Why are copper and aluminium wires usually employed for electricity transmission?**

✔ ANSWER

Copper and aluminium have **very low resistivity**, so they are good conductors.

> Very little energy is lost as heat in them. (Aluminium is also light and cheap.)

🎉 **Chapter 11 done! Revise the answers once more tomorrow.**